

Multi-Bottleneck Fairness for High-Precision Congestion Control

HPCC-FS — one fair-rate field, one scalar per switch port

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HPCC in 60 seconds: precision congestion control

1

Switch stamps raw per-hop state into INT

tx bytes, queue length, link rate — stateless per flow

2

Sender computes exact utilization U

multiplicative update toward target $\eta = 95\%$

3

Near-zero queues, top-tier FCT

beats DCQCN / TIMELY / DCTCP on single-bottleneck benchmarks

the INT header

hop 1: tx, qlen, rate

hop 2: tx, qlen, rate

hop 3: tx, qlen, rate

... up to $maxHop = 5$

The catch: the sender sees load, not its fair share.

8 bytes per hop — the record grows with path length

Modern traffic crosses many contended links in series

Cross-pod ML

allreduce spans aggregation + core
layers

Disaggregation

memory / storage traffic transits
multiple fabrics

Multi-tenancy

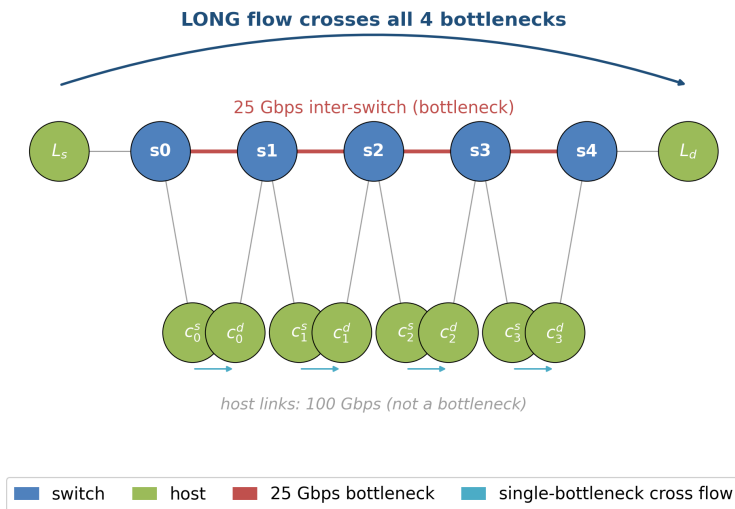
shared links at every tier of the DC

**Does HPCC still deliver fair service when a flow crosses
multiple bottlenecks — not just one?**

Nobody had measured it carefully. We did — deterministically, against a max-min oracle.

The parking lot, scored against a max-min oracle

Parking lot at $N=4$: one LONG flow + one CROSS flow per bottleneck



Setup

N bottlenecks, 25 Gbps each.
One long flow crosses all N ; one cross flow per link.

Fair answer

Every flow gets 12.5 Gbps
(progressive filling, recomputed as flows finish).

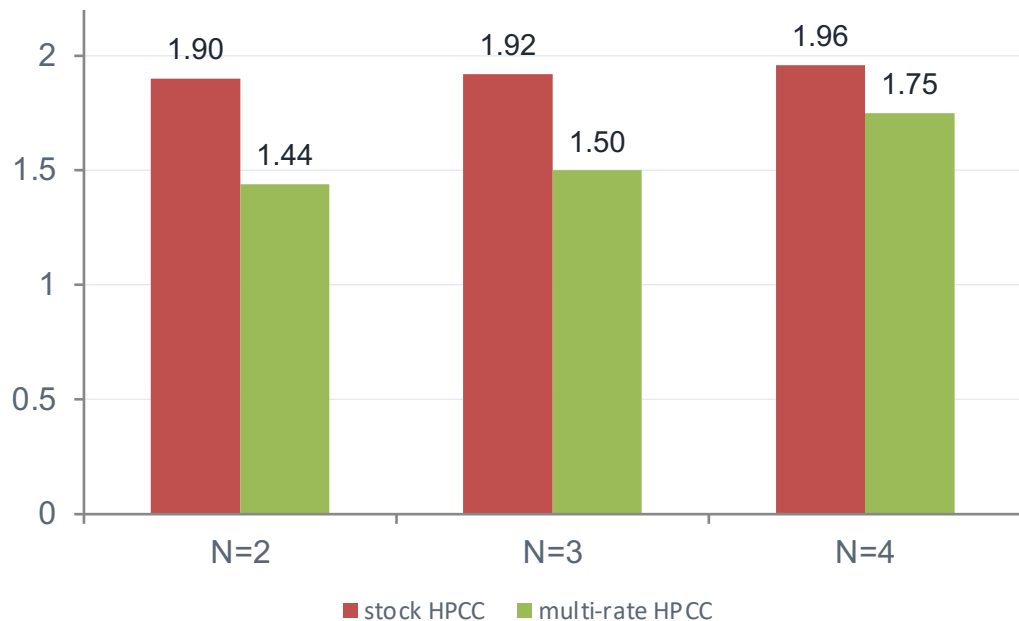
Metric

unfairness = long FCT / mean short FCT. Oracle = 1.0

Deterministic: every configuration reproduces byte-identical artifacts.

FINDING 1

The long flow is starved $\approx 2\times$ — and it's structural



unfairness ratio — max-min oracle = 1.0

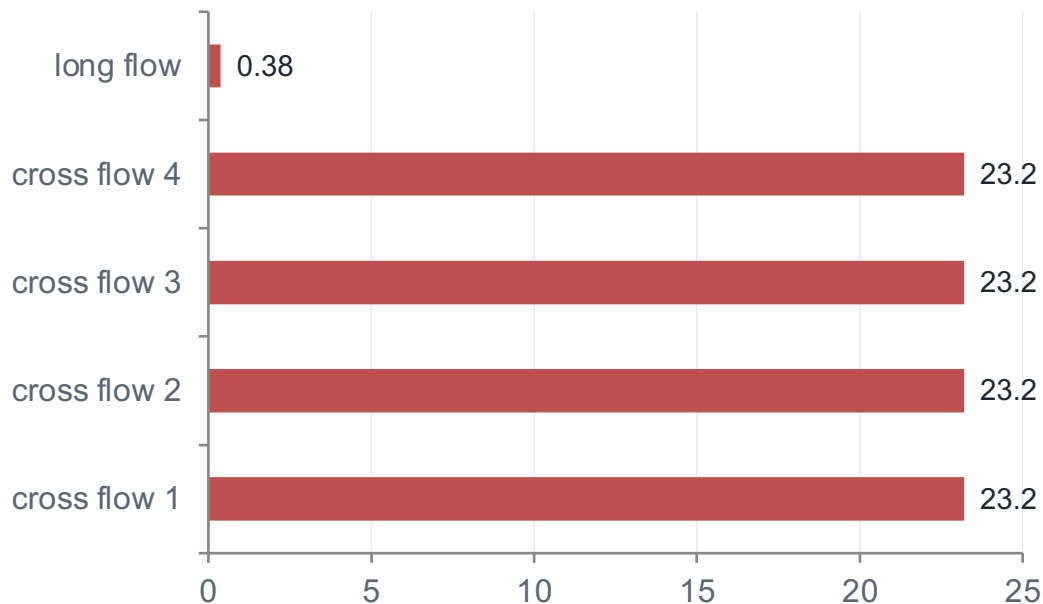
1.96× penalty at N = 4

3.5× when the long flow is small (RPC-scale)

$N \geq 5$ breaks outright: INT maxHop overflow

FINDING 2

Winner-take-all — and nothing is ‘wrong’



stock HPCC, $N = 4$, first 18 ms — fair share is 12.5 Gbps for everyone

Zero PFC. Zero loss.

No pathology to point at — every link is at target utilization η .

The unfair allocation is the control law's equilibrium.

At each shared link the local single-bottleneck flow wins; the long flow is squeezed at all of them.

Five sender-only fixes — none gets close

sender-side variant (mb_mode)	idea	best penalty @ N=4
k-aware additive increase	scale AI by congested-hop count	1.63×
debiased max/mean blend	soften the max-hop bias	1.95×
responsibility-weighted MD	MD scaled by own share	1.89×
k-th-root MD	split the decrease across hops	1.92×
global share-aware AI	AI boosted by (1 – share)	1.71×

Each variant is a 1–2 line change in HPCC's update (*mb_mode 0 stays byte-identical to stock*) — the target is 1.0×

Two structural reasons every sender-only fix stalls

1 • The η -hold equilibrium

Links settle at target utilization $\rightarrow$ HPCC's multiplicative term ≈ 1 .

Every flow holds whatever rate it grabbed at startup. Additive nudges need $\sim 10^4$ RTTs to redistribute.

2 • N is not observable

Fair share is C / N . INT tells the sender C — never N , the link's flow count.

Without N there is no strong multiplicative move toward fair share — only slow AIMD nudging.

The information isn't at the sender — so the fix can't be either. It belongs at the link.

RCP: the link advertises one fair rate to every flow

$$R \leftarrow R \cdot (1 + (\alpha \cdot (C - y) - \beta \cdot q/d) / C)$$

$$\alpha \cdot (C - y)$$

chases spare capacity — link under-full $\Rightarrow$ R rises

$$\beta \cdot q/d$$

brakes on backlog — queue forms $\Rightarrow$ R falls

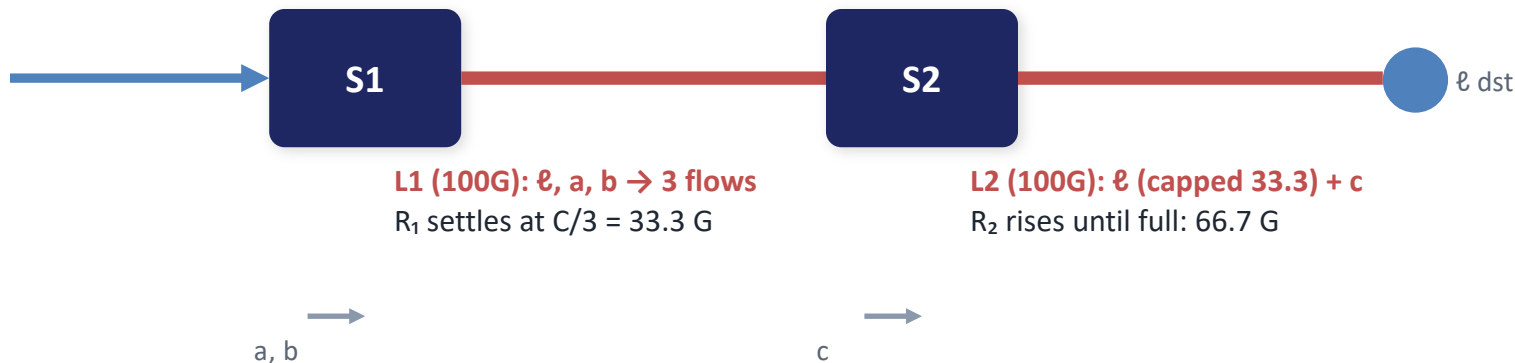
fixed point

$$y = C, q = 0 \Rightarrow N \cdot R = C \Rightarrow R = C/N$$

The link converges to the max-min share *without ever counting flows* — N is implicit in the observed load y.

Worked example: path-min gives network-wide max-min

long flow ℓ



ℓ pkt: fairRate 0 $\rightarrow$ **33.3** $\rightarrow$ $\min(33.3, 66.7) =$ **33.3**

$\ell = a = b = 33.3$

$c = 66.7$

Exactly the max-min allocation — both links 100% full, no per-flow state anywhere.

The whole mechanism: three small pieces

Switch

- one scalar fair rate per egress port
- RCP update once per control interval ($\approx$ RTT)
- state: 3 numbers — no per-flow data

Wire

- new INT mode: one 64-bit fairRate field
- 8 B / packet, independent of hop count
- each hop min-aggregates; receiver echoes

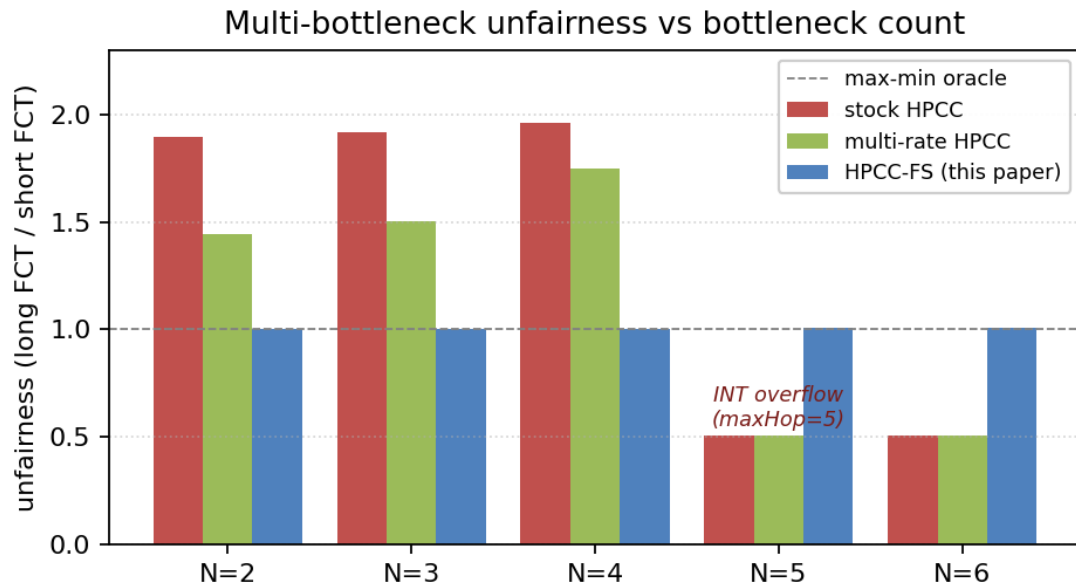
Sender

- new ACK handler: rate $\leftarrow$ path-min fairRate
- clamped to [minRate, NIC]
- rate-only: per-flow window cap off

stock HPCC (cc_mode 3): byte-for-byte unchanged

an operating mode selected per traffic class — not a replacement

Flat at 1.0 across the whole sweep — zero PFC

**1.005×**

penalty at N = 4 (was 1.96×

1.003–1.007×

across N = 2...6, incl. the overflow regime

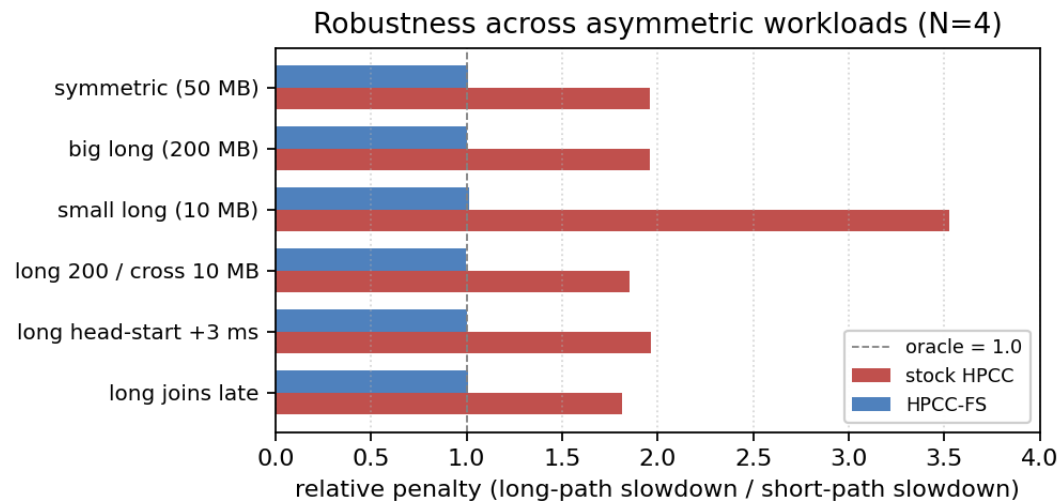
0 PFC

every workload, every N

HPCC-PINT (compressed INT) still shows 1.77–1.88× — a smaller footprint is not the fix; a fair-share signal is.

RESULTS

It generalizes — and holds under asymmetry



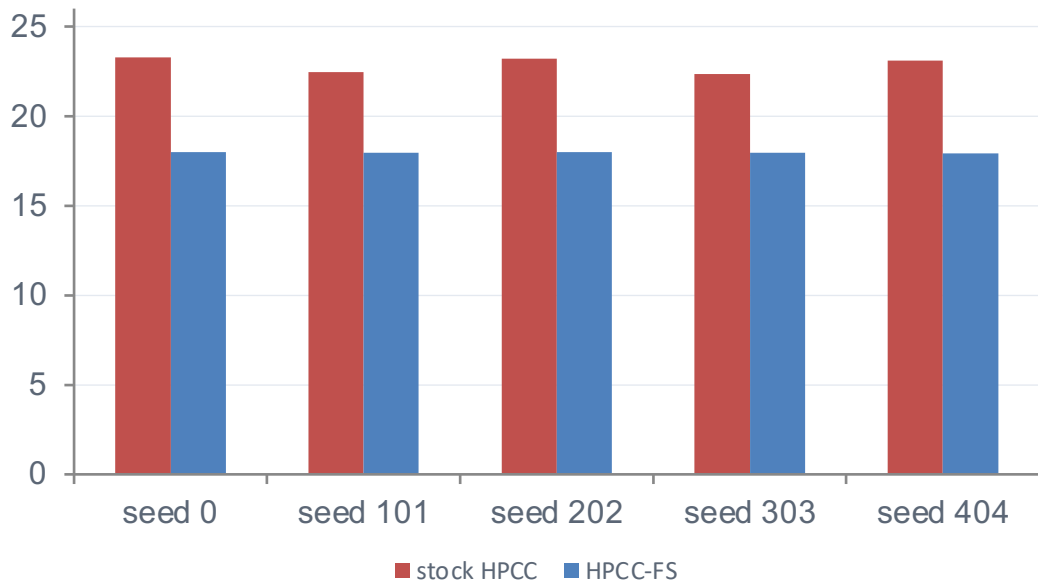
six asymmetric-size / staggered-start variants at $N = 4$: HPCC-FS within 1.0 ± 0.012
(worst stock case $3.53\times \rightarrow 1.012\times$)

topology	stock	HPCC-FS
3-tier tree fabric	1.36×	1.003×
k=4 ECMP fat-tree †	1.34×	1.001×
parking lot N=5,6	breaks	1.006–1.007×

† oracle-normalized (removes ECMP placement luck). Across 5 hash seeds the raw ratio is placement-dominated; HPCC-FS $\leq$ stock at 4 of 5 seeds (the 5th is uncontended).

RESULTS

The job is gated by its slowest flow: -21% JCT



-21%

mean JCT — improves at every seed
(+19.5...+22.7%)

17.9–18.0 ms

FS JCT is placement-invariant (stock 22.3–23.3)

~3× faster

short background flows in this mix — max-min
removes the placement lottery

16-flow ring allreduce, k=4 fat-tree — JCT = slowest flow (ms), 5 ECMP hash seeds

Ablation, sensitivity, and HPCC's home turf

Rate-only is necessary

Re-enable the per-flow window cap
→ 1.5–2.5×, worse with path length.

`var_win` sizes the window from NIC
line rate, not the path — the long flow
gets window-bound below fair.

Parameters not load-bearing

RCP defaults $\alpha = 0.4$, $\beta = 0.226$ —
untuned.

Sweeping α , β , startup fraction,
`minRate` one-at-a-time: penalty stays
1.004–1.008×, PFC = 0 throughout.

Home turf: not a trade-off

Single-bottleneck incast, 8 Gbps
startup:

mean FCT 242 vs 259 μ s
peak queue 38 vs 143 KB

beats HPCC's FCT at $\frac{1}{4}$ the queue —
fairness headline unchanged.

Everything regenerates deterministically from the artifact — figures and tables are built from live runs.

What this is — and what it isn't

1

The mechanism is RCP; HPCC-FS is a mode, not a repair. We claim the diagnosis and the placement. HPCC-FS coexists with stock HPCC per traffic class; mixed classes sharing links are future work.

2

Simulation-only, three topology families. No testbed; no head-to-head with Bolt / PowerTCP (would require re-implementing them). The simulator re-port reproduces the original HPCC baselines end-to-end.

3

Convergence is sub-ms, not one RTT. Within 5% of fair share in ~ 0.6 ms (~ 30 RTTs); path-length-independent.

Future: a hybrid overlay — $\text{rate} = \min(\text{rate_HPCC}, \text{fairRate})$ — keep HPCC's precision and the fairness; needs a path-aware window.

One field, one scalar — a coexisting fairness mode

Diagnosis

HPCC starves multi-bottleneck flows $\sim 2\times$ ($3.5\times$ small flows); breaks past 4 hops. Zero PFC — it's the equilibrium.

Negative result

Five sender-only fixes fail: at η -hold, no strong move exists without the flow count N .

HPCC-FS

A per-class mode alongside HPCC: $1.005\times$, zero PFC, -21% coflow JCT, stock path untouched.

Artifact (code · paper · reproduction · data): github.com/UNOCourseDemo/hpcc-fs

Thank you — questions?

The re-reported simulator reproduces SIGCOMM '19

short-flow p99 slowdown	HPCC	DCQCN	TIMELY	DCTCP	PINT
WebSearch 50% load	1.9	13.7	18.3	5.7	2.1
FB-Hadoop 70% load	6.1	30.6	221.9	8.4	5.0
PFC events (fb 70%)	0	0	1.12 M	0	0

5 schemes × 2 workloads × 3 loads on a 376-node fat-tree (320 servers, 100 G NICs).

All of the paper's signatures reproduce: HPCC's short-flow tail far below DCQCN/TIMELY, sub-MB queues, zero PFC; TIMELY's PFC explosion at load; DCTCP intermediate. Along the way we found and fixed a latent uninitialized-member bug (Node::m_node_type) exposed only at scale.

Mixed-size workloads · the INT-overflow artifact

Mixed sizes (WebSearch dist.)

Long-flow mean slowdown -15%; short-path mean +45%.

This is correct max-min: stock's starvation of the long flow was inflating headroom that short flows enjoyed.

Per-class targets (beyond pure max-min) are application policy — future work.

Why stock 'improves' at $N \geq 5$

The INT header holds $\text{maxHop} = 5$ per-hop records. Longer paths overflow the record; the utilization estimate is computed over the wrong hops.

The $0.51\times$ 'advantage' is corrupted control input, not fairness. HPCC-FS is immune: one field, any path length.